

“Reading Swings”: Giving Fans a Peek Behind the Curtain Of Outfield Defense

Team 312

1. Abstract

Everyone loves a great catch, but a lot of what goes into making great catches can't be seen in a short highlight. In reality, the outfielders' efforts begin before the pitch is even thrown – their chosen starting point and how they use information from the hitter to adjust from there has an unseen impact. This paper evaluates this skill, which is called “reading swings”, and introduces a scoreboard graphic so fans can appreciate it mid-game. For each fly ball, we compare the real catch probability to what it would've been if the fielder was in the same position he started the at-bat, and show how much easier he made the catch by moving. This highlights a skill left out of current defensive metrics like [OAA](#), giving fans a greater appreciation for outfielders, and teams the means to evaluate something they previously couldn't.

2. Background

Before getting into the details of the previous work this paper builds on and the development of the graphic to display the concept, it's important to establish exactly what we mean by “reading swings”. An important thing to note is that we use “reading swings” here as a term of art, as our dataset does not include swing data. It's well documented that teams are always trying to predict where opposing hitters will hit the ball. Pro teams have used ball tracking data to create spray charts for each hitter and models to determine where each fielder should stand before the game.

Players have been doing this on the field on a pitch-by-pitch basis since long before the creation of ball tracking technology by reading the hitter's response to previous pitches to predict whether they are swinging late, early, or on-time and moving a few steps to adjust to this new information. For example, a left fielder playing near the gap that sees a right-handed hitter pull a fastball foul will likely move a few steps right, expecting the hitter to continue to swing early. When outfielders

make these adjustments correctly it can save precious seconds by shortening the distance to the eventual spot of the catch. That extra opportunity time makes the difference between outs and runs. The purpose of our graphic is to show the value of these adjustments, engaging fans with a subtle nuance of the game and giving them a better appreciation for outfield defense while giving teams the means to visualize who is most effective at this skill.



Figure 2: Dodgers outfield playing a lefty hitter to pull, with the right and centre fielders shaded towards the first-base foul line while the infield is in the now-banned defensive shift.

Outfield defense has been a topic of extensive research and discussion for years and while there hasn't been any research on "reading swings" directly, this paper builds on years of research in connected concepts, primarily player range, route efficiency, and optimal defensive positioning. Though positioning research has mostly focused on infielders with the rise and eventual ban of the infield shift in 2023, outfield defensive positioning has still been researched heavily. So much so that almost every MLB team gives their outfielders cards on the field that determine their starting position before each at-bat. One noteworthy study from 2020, [Swing Shift: A Mathematical Approach to Defensive Positioning in Baseball](#), by Bouzarth et al. used mathematical programming to model optimal defensive positioning by ignoring traditional positions to minimize opponent BABIP. They used the hitter's tendencies and strategy to position fielders, penalizing for leaving too much of the field open to

significantly lower expected BABIP compared to standard shifts. There has been other research on optimizing positioning, but this paper delves into how that optimal position changes as the at-bat progresses and visualizes how impactful knowing that change can be.

Additionally, though there is no directly comparable research in baseball, there is in other areas. Poker researchers have attempted to solve a very similar problem, the paper [Baye's Bluff: Opponent Modelling in Poker](#) by Southey et al. fits a probabilistic model that takes the prior knowledge they have of the opponents' strategy and updates to a posterior belief after each action the opponent takes in the hand. This is directly comparable to "reading swings" in baseball. Every pitch in the at-bat is like a card dealt in a hand and each swing is the opponent's reaction, and where the ball will be hit is the opponent's "strategy". The fielder's initial position is the prior belief, and after each swing, the fielder moves to his posterior belief of where the ball will be hit. We aren't solving this problem directly though, we're creating a method to measure success at solving this problem among outfielders and using a scoreboard graphic to highlight those who do it best.

3. The Data

To create this graphic, we first need to understand the data we have and filter it down to just the relevant plays. The dataset in question contains minor league play-by-play and player tracking data from which we will take a full season. Unfortunately, since this dataset doesn't include swing tracking data, we can't measure how effective a player's process of "reading swings" is, only the result of their adjustment.

We're working with outfielders, so we need to filter down to fly balls hit to the outfield. Then, since the data has been anonymized, we'll keep it to fair balls that were either caught or bounced first on the field to avoid complications with wall and stadium dimension data, as that's not included in the dataset. Since this paper focuses on "reading swings", we further filter this set of flyballs to those not hit on the first pitch of the at-bat. This gives us a distinct starting point prior to the first pitch of

the at bat and an ending point prior to the pitch that is hit. For each relevant fly ball, we have the fielder's position at the time of the pitch, the trajectory and final position of the batted ball, its hang time, opportunity time, and the distance the fielder covered or needed to cover. This is all we need to create our catch probability model.

4. Methodology

The bedrock of this project is the catch probability model, as that is the basis on which we evaluate the effectiveness of the fielder "reading swings." It is important to note that a play was labeled a catch only if the responsible outfielder acquired the ball before it bounced. However, if it touched the ground first, the play was labeled a non-catch, even if the ball was acquired subsequently.

After going through the aforementioned filtering process, the next step was to identify what factors are most important to catch probability. The first and most obvious one is opportunity time, which is measured as the number of seconds from the pitch to when the ball reaches the catch point and represents how much time the fielder has to make the play.

Next, we calculate the distance the fielder has to cover, which is measured as the straight-line distance from the fielder's starting position to either the catch point or the point of the first bounce, depending on whether the ball was caught or dropped for a hit. Since any of the three outfielders (left, center, or right) may be responsible for any given fly ball, we assign it to the outfielder who began closest to the ball's landing spot, rather than assuming a fixed fielder (such as a center fielder) will field every ball.

The last thing we took into account was the route angle as compared to home plate. It's well known that going back on a ball is much more difficult than coming in, and in most cases, the closer to directly back the angle is, the more difficult the catch becomes. To incorporate this, we created a function that finds the angle at which the fielder is running and applies a catch probability "penalty" to plays where the fielder

is running at a 90° - 270° (exactly perpendicular or backwards) angle from home plate, with the largest impact being at exactly 180° . Due to the anonymization of the data, walls and stadium dimensions could not be included for this paper, but should be in future analyses.

Now that our factors have been defined, the next step is to use them to accurately model catch probability. We did this with a logistic regression model with 1% accuracy, see Appendix B for more details on the model itself and the model selection process.

Finally, to complete the process required to create the scoreboard graphic, we used our model to compare the actual catch probability of each relevant fly ball to what it would have been had the fielder never adjusted his starting point from the beginning of the at-bat. To do this, we took the calculation for each fly ball and simply switched the starting point inputs to what they were before the first pitch of the at-bat was thrown and kept everything else the same to get our hypothetical catch probability. Then, to create the final product, we took the two starting points and used them to illustrate the different routes to the ball, overlaid them into a single visual, and displayed the catch probability difference between the two routes.

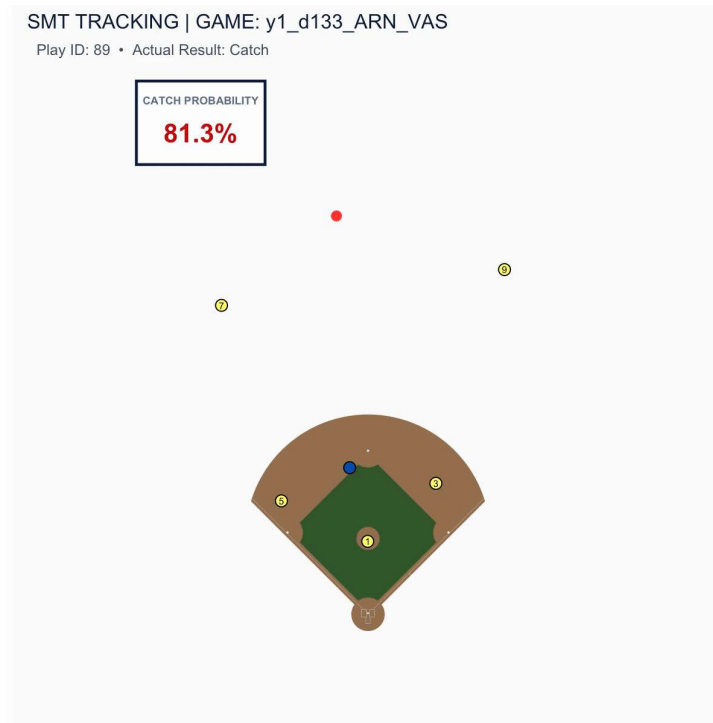


Figure 3: Graphical representation of a play made by the centerfielder. This catch had a 81.3% catch probability and resulted in a catch. The graphic shows the centerfielder moving back and to the right to make the play, with a line stretching out from their original position to where the play was made.

<https://drive.google.com/file/d/1fL-DEDC5FF7P3d1ZSgXXU9ESFOByqrda/view?usp=sharing>

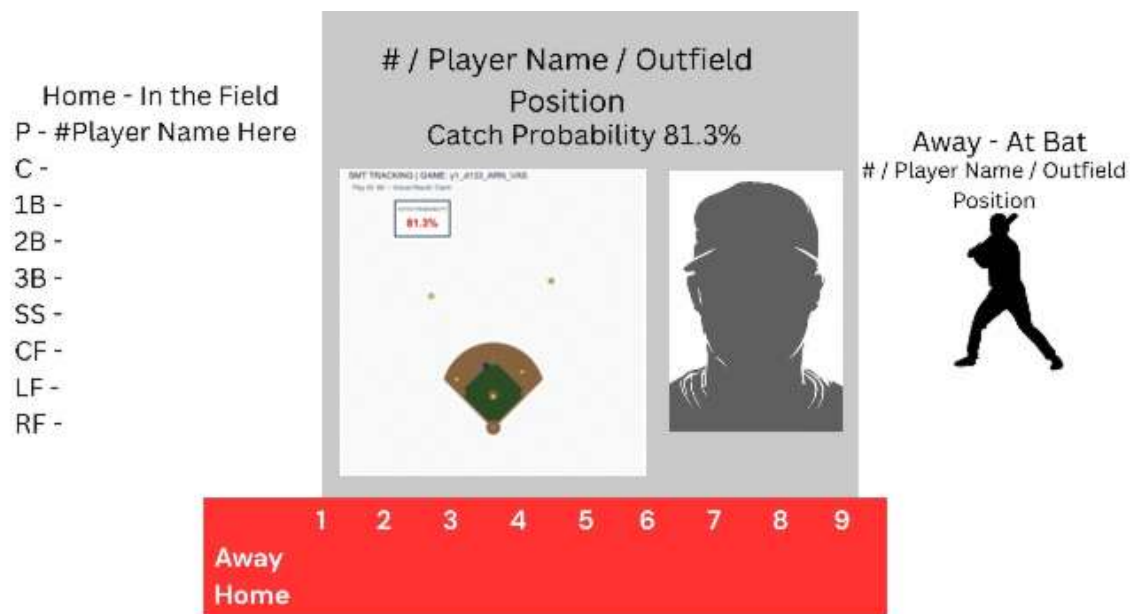


Figure 4: Mock-up scoreboard displays the graphic created with catch probability.

5. Implementation

When deployed effectively, this graphic can be a great tool that generates fan engagement with the nuances of the game. The best time to use it is after a highlight reel catch that was only made because the outfielder made the correct mid-at-bat adjustment. Displaying the importance of that level of detail in the professional level like that gives the casual fans in attendance a whole new aspect of the game to engage with and think about. Another way to highlight this is later in the game, between innings to spotlight a particular outfielder who made several correct adjustments in that game that paid off. This graphic, coupled with highlights of the catches they led to, would allow the fans to appreciate outfield defense on a deeper level and give them something interesting to look out for between pitches.

The graphic shown above is just a sample of what would go on a scoreboard that is constrained by the limits of the data we had. In an actual stadium with live play-by-play data and video, the graphic would feature the outfielder's name and incorporate the dimensions of the specific field. Further team-specific animations could also be included to create more fan engagement and allow the graphic to blend in with the rest of the game presentation.

6. Conclusion

There's a lot more that goes into making great plays in the outfield than what most fans know to pay attention to during the game. The concept of "reading swings" is something those in the game understand, but casual fans are often unaware of despite its importance to effective outfield defense. Showing new elements of the game to fans and giving them a deeper appreciation for the nuances and added layers of complexity between pitches in an engaging way is a great step to capturing new lifelong fans.

This is an area that hasn't been researched much in the past, and the dataset's limitations along with the paper's constraints allow for many fascinating ways to build off this project. This could mean further developing the graphic to include stadium-specific data for greater accuracy and fan engagement. Or the analysis could be deepened using modern bat-tracking, pitch type, and catcher stance data to evaluate the process with which outfielders "read swings", studying how the implementation of PitchCom has impacted the skill, applying the same concepts to infielders while taking into account the base-out state and team strategy for each play. Like almost any other skill, this can also be used for player evaluation: following the same model as fielding runs based on OAA, one could take the cumulative catch probability improvements the player creates by effectively "reading swings" and convert that value to outs saved and, in turn, runs saved.

7. Acknowledgements

This project would not have been possible without help from other resources around the baseball, coding, and data analysis community. Firstly, this [Fangraphs article](#) detailing the process behind an early catch probability model, as well as another [article from Medium](#) by Bryce Gayan showing how he created a model for NCAA baseball. Finally, on the coding side, we'd like to acknowledge the R help we got from the [sportyr package and sportsdataverse](#) and from this [Stack Overflow post](#).

8. References

- Bouzarth, E., Harris, J., & Grannan, B. (2020, September 15). *Swing shift: A mathematical approach to defensive positioning in baseball*.
<https://www.degruyterbrill.com/document/doi/10.1515/jqas-2020-0027/html?lang=en>
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<https://www.mlb.com/glossary/statcast/outs-above-average>

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9. Appendix A: Detailed Data-handling

order of appendices and exact headings still to be finalized

A.1 Filtering Methods

Plays were first filtered to exclude the first pitch, as only the subsequent pitches would reflect adjustments made by fielders based on observing prior swings. Then, plays would be included only if the event of the ball being hit into play (event code 4) where the event immediately after was not a catch, home run, or foul. Next, the hang time calculated required the timestamp of the event itself and the next event, so any data with no valid next event was filtered out. Finally, based on calibrating threshold through several runs, balls that didn't meet a specific threshold for hit distance were excluded to avoid involving any weak contact or bunts that were extremely unlikely to reach outfielders.

A.2 Hit Distance Threshold

As stated above, the hit distance threshold value of 21 was selected as a result of observing the distribution of hit_distance values in the particular portion of the dataset being used. This threshold allowed for a usable sample of outfield fly balls that excluded contact too weak to represent a true opportunity for an outfielder.

A.3 Feature Definitions

For each fly ball, the following features were calculated in regards to the outfielder responsible for the play.

1. Initial Distance: Calculated the straight-line distance between the fielder's starting position and the ball's position at the moment of contact.
2. Initial Angle: Utilized R's atan2 function to calculate the angle (degrees 0-360) between the fielder's starting position and the ball's position at contact.

3. Opportunity Time: Calculated the elapsed time, in seconds, between the pitch and the ball reaching a touch point, either an acquisition or a bounce (ball event codes 2 and 16, respectively).
4. Initial Speed: Calculated the fielder's average speed over the first three position changes at the start of the play to measure reaction quickness.

10. Appendix B: Expanded Modelling

B.1 Model Specification

Formula/Variables Used: The likelihood of a fielder making a catch (`catch_status`) based on the constraints: initial distance, initial angle, initial speed, and opportunity time.

`catch_status ~ initial distance + initial_angle + initial_speed + opportunity_time`

1. Logistic Regression - `glmnet`
2. Random Forest - `ranger`

Train/test split: The data was partitioned 80% training and 20% testing. This choice of split was based on the catch/non-catch outcome to specifically ensure both partitions preserved a representative ratio of the classes. This was important because of the moderate sample size, since a random split would risk producing a test set with a different class balance than the training set and would distort evaluation metrics.

Catch probability is something that has been modelled using several different methods in recent baseball research, so there was a lot to build on when creating our model. The formula just amounted to summing up the factors such that:

`catch_status ~ initial distance + initial_angle + initial_speed + opportunity_time`

For our purposes, though, we still wanted to test a couple of model types to see which would be most effective: a logistic regression model and a random forest model. The former used the “`glmnet`” R package with a penalty of 0.01 and a mixture of 1. For the random forest model, we grew 500 trees and used the “`ranger`”

package. To test these two models, we used an 80%-20% training-testing split on 21 plays, then evaluated each model on test accuracy and compared their predicted probability spreads to determine that the logistic regression model was better compared with the random forest model.

B.2 Graphic Creation

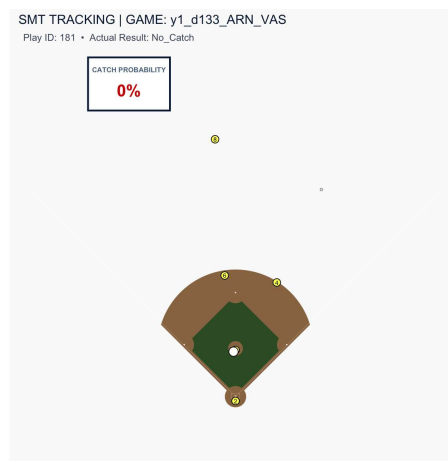
To create the animated graphic, we utilized the `sportyR`, `gganimate`, `gifski`, `dplyr`, and `ggplot2` functions to demonstrate how the outfielders read swings to make a play.

Utilizing player position and ball position data at the same timestamps

The final pipeline to animate the GIFs consisted of three core functions that were encompassed in the final GIF-creating loop. First, `build_tracking_data()` gathered all the raw animation data, including tracking the frame rate based on player positions, then anchoring the animation to the timestamp of the pitch, then labeling the ball tracking layer, followed by labeling every tracked player (color-coded), and finally combining to a single data frame. This final result makes it ready for the next plotting step for `gganimate`.

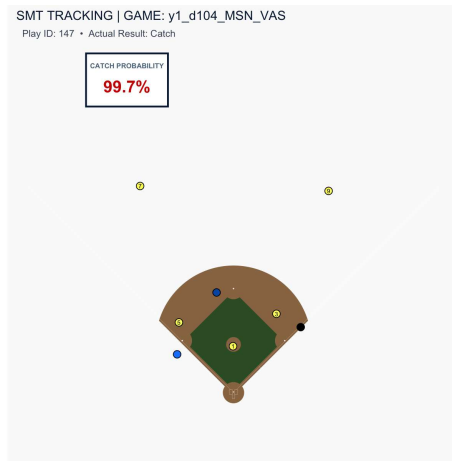
Following this, `generate_play_gif()` accomplishes other core functionality. It predicts the catch probability using the previous model and displays it as a percentage, determines the end of the GIF as the timestamp when the catch occurs, then builds the tracking data using the earlier function. After this, `sportyR` is used to create the base layer of the graphic, then player and ball positions are added as points, with `transition_time()` converting the plot into an animation. This whole process is looped and applied to all the plays after the filtration process.

The graphic can display differences in catch probability, even showing ground balls that weren't possible for outfielders to catch. For example, this graphic displays 0% catch probability and a bounce can be seen as it approaches the right fielder. Although this play would not be one that teams would want to display on scoreboards, it can help visualize ball movement.



<https://drive.google.com/file/d/1nDHBW2WcxmpiNflgKwfbb8-CLYewjiiU/view?usp=sharing>

In contrast, this graphic displays a play with a 99.7% catch probability, with the result of a catch by the center fielder, and the ball can be seen going straight to CF as the fielder backs up.



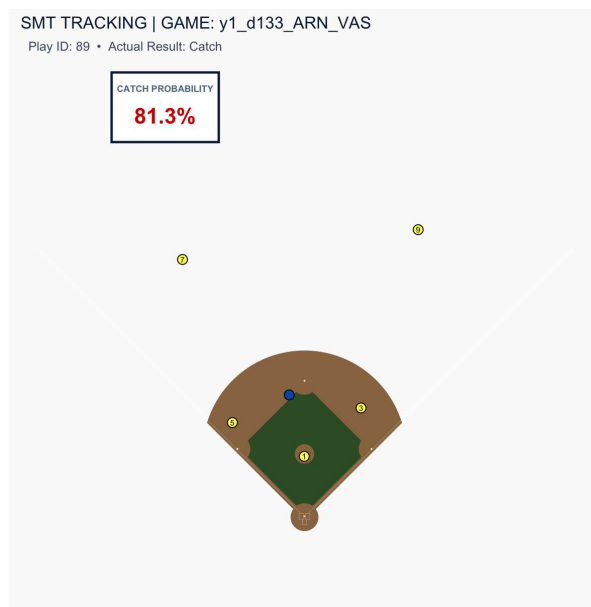
<https://drive.google.com/file/d/14nJ26lq0tSrnkT9eld0Ye9rpoXNicCK8/view?usp=sharing>

The differences between almost guaranteed catches and improbable catches can be viewed in the graphic. Ideally, this graphic would have limited player details to only highlight the player who made the catch, along with a clearer track of their movement throughout the at-bat.

B.3 Future Steps for the Graphic

The graphic created could be improved by having a "hypothetical" player modeled that would track the path the outfielder moved throughout the play. This "hypothetical player" would start at the spot where the actual fielder was standing during the first pitch of the at-bat and move in a straight line towards the eventual landing point of the ball at the same time as the real fielder. The following graphic is what we were able to create with our model. In a future mid-game, fan engagement setting, we would use instant replay data to capture the exact spot of the fielder at the start of the at-bat and track exactly the steps he takes after each pitch by “reading swings”

and show how much that improved his ability to make a play on the batted ball.



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B.4 Graphic Implementation

Figure 4* displays a mock scoreboard with the created graphic displayed. This is to help visualize how the graphic could be implemented in-game. As mentioned in section 5, Implementation, this graphic could be displayed in between innings after an outfielder makes a good play. Alongside the replay of the catch, the graphic would help demonstrate the impact of the player's movement. Displayed with the player's information and the catch probability percentage, fans would be able to get an insight into the defensive skills of the outfielders. In this mock scoreboard, the graphic is displayed next to the player who made the play, only taking up part of the scoreboard. With more data and individual team graphic aesthetics, ideally teams could cut away from a split screen and display only the graphic of the play to draw more attention to it, especially when it was a harder play with a lower catch probability.